Environmental Progress Report



Covering fiscal year 2023

Contents

Introduction

3 Reflections from Lisa Jackson
4 Report highlights
5 Goals and progress

Environmental Initiatives

Apple 2030

11 Journey to Apple 2030
12 Approach
15 Design and materials
24 Electricity
32 Direct emissions
35 Carbon removal

Resources

39 Approach
40 Product longevity
45 Material recovery
48 Water
52 Zero waste

Smarter Chemistry

58 Approach
59 Mapping
61 Assessment
63 Innovation

Engagement and Advocacy

67 Approach
68 Listening to a range of voices
69 Achieving change together
73 Supporting communities worldwide

Data

77 Greenhouse gas emissions
78 High quality carbon certificates
79 Carbon footprint by product
81 Energy
82 Resources
83 Normalizing factors

Appendix

85 A: Corporate facilities energy supplement
94 B: Apple's life cycle assessment methodology
96 C: Assurance and review statements
107 D: Environment, Health and Safety Policy
109 E: ISO 14001 certification
111 Report notes
112 End notes



Reflections

Apple's commitment to climate action has never been more clear.

Thanks to the hard work of teams across our company, we've reduced our emissions by over 55 percent since 2015. We've crossed key milestones on our environmental journey. And we're rising to the generational challenge of climate change by working with companies and communities that span the globe.

Apple 2030 is our commitment to be carbon neutral for our entire footprint by the end of the decade. We'll get there by innovating at every stage of the product lifecycle — from how they're made, to what they're made from.

That starts with bringing new clean energy online across our supply chain. Today, more than 320 suppliers have committed to using renewable electricity for Apple production. With over 16 gigawatts already online, they're avoiding more than 18 million metric tons of greenhouse gas emissions.

Our teams and suppliers are a part of so many communities, so Apple's work to protect the planet takes us around the world.

That's why we're partnering directly with communities and local organizations to support environmental solutions where they're needed most. In 2023, that meant expanding our Power for Impact program to bring clean energy to underserved communities in Nepal and Colombia, and supporting programs that increase access to safe water and sanitation in India.

We're also building our products with more recycled and renewable material than ever — work that helps to drive down our carbon footprint. Our customers play an important role in this effort. When you participate in the Apple Trade-In program, you're helping to bring us closer to the day our products will be made without taking anything from the Earth.

The proof is in our products. Last year, more than 20 percent of the materials we shipped in Apple products came from recycled sources. MacBook Air is our first product made with more than 50 percent recycled material, and we're making important strides across all of our products. The iPhone 15 lineup, Apple Watch Series 9, and Apple Watch Ultra 2 use 100 percent recycled cobalt in their batteries. More than 99 percent of the tungsten in our products comes from recycled sources. And we're driving innovation around the world to improve the way our industry recycles and recovers materials.

We're also finding new and better ways to get Apple products into people's hands. It's not just trains, planes, and automobiles — increasingly, we're shipping Apple products on the open seas, because ocean freight can reduce emissions by as much as 95 percent compared to air transport. And we continue to restore our planet's ecosystems — like the Atlantic Forest, which stretches from Brazil to Paraguay — that play a key role in removing carbon from the atmosphere.

What does all this work add up to? Progress. And that progress is sorely needed, because the impacts of climate change are all around us. Its frontlines are not in boardrooms or government buildings — they're in communities. And we owe it to our global community to rise to the challenge of climate change with all the innovation, empathy, and commitment we can muster.

Lisa Jackson
VP, Environment, Policy and Social Initiatives



Report highlights

Reduced our overall emissions by more than 55 percent

We reduced our overall greenhouse gas emissions across scopes 1, 2, and 3 by more than 55 percent compared with our 2015 baseline year — not including offsets.[1] In that same time period, revenue grew by 64 percent. We estimate that we've avoided 31 million metric tons of emissions through reduction efforts like transitioning our supply chain to renewable electricity and sourcing recycled content.

Read more on page 14.

Introduced our most significant product emissions reductions to-date

Innovations in design and clean energy drove dramatic reductions in greenhouse gas emissions for the fall Apple Watch lineup — the most significant from our business-as-usual scenario.[2] This included 100 percent clean energy for manufacturing and product use, 30 percent recycled and renewable material by weight, and 50 percent shipping without the use of air transportation.[3]

Read more on page 9.



Advocated for reporting transparency

As part of our commitment to greenhouse gas emissions disclosures, we endorsed the landmark California Climate Corporate Data Accountability Act (CA SB 253) to improve transparency and drive progress in the fight against climate change.

Read more on page 71.

Increased our use of recycled materials

We're making steady progress on our journey toward using only recycled and renewable materials in our products. In 2023, 22 percent of the materials we shipped in Apple products came from recycled sources.[4] This includes, 99 percent of tungsten, 71 percent of aluminum, 52 percent of cobalt, 25 percent of gold, and 24 percent of lithium in our products.[5]

Read more on page 17.



MacBook Air is the first Apple product made with 50 percent recycled content

This includes 100 percent recycled aluminum in the enclosure, 100 percent recycled rare earth elements in all magnets and, in another first for Apple, 100 percent recycled copper in the main logic board.

Read more on page 16.

More than 320 suppliers committed to renewable electricity

As of March 2024, more than 320 suppliers have committed to sourcing renewable electricity for Apple production, representing 95 percent of our direct supplier spend. We've accelerated progress with a mandate in our Supplier Code of Conduct for all direct suppliers to transition to renewable energy in the manufacturing of Apple products.

Read more on page 27.



Launched Grid Forecast

In the contiguous United States, we introduced Grid Forecast, a new tool in the Home app on Apple devices. This tool shows customers when cleaner electricity is available from the grid, empowering them with information to help reduce greenhouse gas emissions.

Read more on page 30.

Goals and progress

Emissions

Goals	Progress	Highlights
Become carbon neutral for our corporate operations.	Achieved	We achieved carbon neutrality in 2020 by expanding energy efficiency initiatives, sourcing 100 percent renewable electricity for Apple facilities, and investing in high-quality carbon credits to offset the remaining hard-to-decarbonize corporate emissions.
Achieve carbon neutrality for our entire carbon footprint by 2030 — reducing related emissions by 75 percent compared with 2015.[6]	Ongoing	Since 2015, we've reduced emissions by over 55 percent across our value chain.
Transition our entire value chain, including manufacturing and product use, to 100 percent clean electricity by 2030.[7]	Ongoing	As of March 2024, more than 320 suppliers have committed to use 100 renewable energy for their Apple production. These suppliers account for 95 percent of Apple's direct spend for materials, manufacturing, and assembly of our products worldwide.[8]

Materials

Goals	Progress	Highlight
Use only recycled and renewable materials in our products and packaging, and enhance material recovery.	Ongoing	In 2023, 22 percent of the material we shipped in our products came from recycled or renewable sources, and we also added titanium to our priority materials list in 2023.[9]
Transition to 100 percent recycled cobalt, tin, gold, and rare earth elements in select components and applications by 2025.[10]	Ongoing	In 2023, 56 percent of the cobalt shipped in Apple-designed batteries came from recycled sources.[11]
Remove plastics from our packaging by 2025.[12]	Ongoing	Our product packaging shipped in 2023 contains only 3 percent plastic — down from 21 percent in 2015.[13]

Resources

Goals	Progress	Highlight
Replenish all our corporate freshwater withdrawals in high-stress locations by 2030.[14]	Ongoing	We've partnered on freshwater replenishment projects resulting in 31.2 million gallons of volumetric water benefits.
Certify all Apple-owned data centers to the Alliance for Water Stewardship (AWS) Standard by 2025.[15]	Ongoing	We've certified five data centers since 2021, and 20 suppliers since 2017 to the AWS Standard.
Expand and grow supplier participation in the Supplier Clean Water Program, prioritizing high water stress locations and driving participants to an average 50 percent water reuse rate by 2030.[16]	New	Through our program, we've supported an average 42 percent reuse rate across our 242 participating supplier facilities.
Eliminate waste sent to landfill from our corporate facilities and our suppliers.	Ongoing	Our corporate facilities waste diversion rate increased to 74 percent, driven by progress at our data centers. Throughout 2023, 100 percent of established final assembly sites maintained zero-waste-to-landfill operations.

Overview
Apple 2030
Journey to Apple 2030
Approach
Design and materials
Electricity
Direct emissions
Carbon removal
Resources
Approach
Product longevity
Material recovery
Water
Zero waste
Smarter Chemistry
Approach
Mapping
Assessment
Innovation

Environmental Initiatives



Focus on the future
More than 38 percent of manufacturing electricity for iPhone 15 Pro and iPhone 15 Pro Max is sourced from our supplier clean energy projects.

Environmental Initiatives

Apple 2030

Apple 2030 is our commitment to be carbon neutral for our entire carbon footprint. Our journey to 2030 is focused on first reducing our scope 1, 2, and 3 greenhouse gas emissions by 75 percent compared with 2015, and investing in high-quality carbon removal solutions for the remaining emissions.

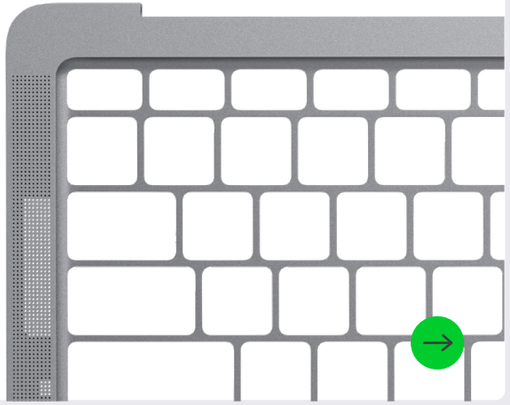
Design and materials
Electricity[17]
Direct emissions
Carbon removal



Resources

We aim to make durable, long-lasting products and enhance material recovery. And we're committed to stewarding water resources and eliminating waste sent to landfills.

Product longevity
Material recovery
Water
Zero waste



Smarter Chemistry

Through chemistry innovation and material selection, we design our products to be safer for anyone who assembles, uses, or recycles them — and to be better for the environment.

Mapping
Assessment
Innovation



The time for change
The significant emissions reductions associated with the Apple Watch lineup mark a milestone toward our 2030 goal.



Apple 2030

An ambitious goal for 2030: We committed to be carbon neutral for our entire carbon footprint by the end of the decade.

Our journey to 2030 centers on reducing our scope 1, 2, and 3 emissions first — upstream and downstream — by 75 percent before balancing remaining emissions with high-quality carbon removals.

We've already reduced emissions by more than 55 percent since 2015, even as revenue grew by 64 percent during that same eight-year period.

We're focused on decarbonizing the three largest sources of emissions — materials, electricity, and transportation — across our value chain. We're prioritizing efforts to drastically reduce these emissions with initiatives focused on achieving the broadest impact before applying offsets. For emissions that can't be further reduced, we primarily look for high-quality, nature-based credits. We invest in carbon sequestration and removal projects to help us achieve carbon neutrality.

Decarbonizing our supply chain is essential to achieving Apple 2030. We've focused on innovative approaches to significantly expand renewable energy across our supply chain and manufacture our products with recycled and renewable materials.



Decarbonizing our value chain

Electricity
Electricity for manufacturing and charging devices represents the largest source of Apple's emissions across all product lines. Achieving carbon neutrality for our entire carbon footprint requires a transition to 100 percent clean energy and significant energy reductions throughout our supplier facilities.

We launched the Supplier Clean Energy Program in 2015 to advance renewable energy throughout our manufacturing supply chain. As of March 2024, more than 320 global suppliers — representing 95 percent of Apple's direct manufacturing spend — have joined the program. These suppliers have committed to using 100 percent renewable electricity for all their Apple production by the end of this decade. The success of this initiative has led to the next phase of our efforts toward achieving a carbon neutral supply chain by 2030.

To rapidly scale and accelerate the progress of the Supplier Clean Energy Program, Apple updated the Supplier Code of Conduct to require all direct suppliers to transition to renewable energy in the manufacturing of Apple products. The codification of this requirement is the next step in our decarbonization journey and a sign of our commitment to integrate supply chain decarbonization into our core business decision-making. We hope that our approach can serve as a model that other companies can follow.

We're also working to address emissions from product use through investments in renewable energy so that every watt of electricity associated with our customers' product use is matched by clean electricity. We're prioritizing product efficiency to reduce the electricity use of our products.

We've also introduced Grid Forecast, a tool in the Home app that informs U.S.-based users when their power grid has cleaner energy available for use. And with iOS 16, we launched the Clean Energy Charging feature in the United States, which looks at the sources of the electricity during expected charge times and optimizes for when the grid is using cleaner energy sources like solar or wind.

Materials
We've advanced toward our 2030 goal by using recycled and renewable materials, which often have a lower carbon footprint than primary materials. And we announced that, by 2025, we plan to use 100 percent recycled cobalt in all Apple-designed batteries, 100 percent recycled tin soldering, 100 percent recycled gold plating in all Apple-designed rigid and flexible printed circuit boards, and 100 percent recycled rare earth elements in all magnets across new products.[18]

We've pioneered the use of many recycled materials in our products through world-class product engineering, extensive design qualifications, and supply chain engagement. Our focus is on materials that will yield significant environmental and social impacts, even when those materials come with notable challenges to building circular supply chains. To read more about our efforts to create circular supply chains, see the Prioritizing our efforts section. We intentionally design products to minimize the use of primary (non-recycled) materials and maximize recycled content as feasible. And year over year, we innovate and improve manufacturing processes to use materials more efficiently and decrease manufacturing scrap.

In 2023, manufacturing our products accounted for 59 percent of our gross carbon footprint. This includes emissions from fuel combustion; heating, ventilation, and air conditioning (HVAC); refrigeration; use of fluorinated gases; and other physical or chemical processes (excluding transportation). These emissions sources vary widely and require use of diverse technologies and solutions to abate.

Our approach is to identify process emissions across our supply chain, like the use of fluorinated gases in display and semiconductor manufacturing, and launch targeted programs to address these emissions in partnership with our suppliers, governments, and industry stakeholders. We continue to launch supplier programs targeting emissions from manufacturing operations and facilities used to produce Apple products, like the Supplier Energy Efficiency Program launched in 2015 to help our suppliers optimize their energy use.

Transportation
In 2023, transporting our products to customers accounted for 9 percent of our gross carbon footprint. To address the emissions from transportation, we're shifting more product volume to shipping modes that are less carbon intensive than air transport, such as ocean or rail. Shipping the same Apple product by ocean generates 95 percent fewer emissions than shipping it by air, based on our carbon footprint methodology. We're investigating a transition to low-carbon sustainable aviation fuels (SAF) to reduce the carbon footprint of air shipment. We're also addressing transportation-related emissions through product and packaging design that drives reduced packaging mass and volume, creating boxes that use space more efficiently.

Carbon removals
Prioritizing emissions reductions across our entire value chain is critical to limiting climate change to 1.5°C. Still, not all emissions can be avoided or reduced with existing solutions. And some existing solutions will require greater industry and government efforts to scale before broader commercial adoption becomes feasible. As we focus on eliminating the majority of emissions across the value chain, we're also addressing emissions we can't reduce by looking for high-quality carbon credits from nature-based projects. The nature-based projects we select focus on carbon sequestration, like planting forests and restoring mangroves, and have additional benefits that improve climate adaptation and resilience.

Transparency and integrity are critical to the carbon projects that we fund. The credits from these investments must be additional, permanent, measurable, and quantified — with systems in place to avoid double-counting — and must avoid leakage. To read more about our work in carbon removals, see Carbon removal and read our white paper, Apple's Carbon Removal Strategy.

Journey to Apple 2030

Where we've been

2015

Launched
Supplier Clean Energy and Energy Efficiency Programs established to partner with global suppliers to reduce the energy used to manufacture Apple products

2018

Achieved
100% renewable electricity
generated or sourced for our corporate operations

2020

Achieved
Carbon neutrality
for our corporate operations and announced 2030 targets

2021

Launched
Restore Fund
to advance nature-based carbon removal projects

2022

2023 activities

22%
of the material shipped in products came from recycled or renewable sources[19]

20%
reduction in emissions from transporting products compared to 2022

320+
suppliers are committed to using 100% renewable energy for all Apple-related production by 2030

12.8M
devices and accessories sent to new users for reuse from programs like Apple Trade In and AppleCare

70%+
reduction in average product energy use since 2008

Codified
100% renewable energy
mandate added to our Supplier Code of Conduct applicable to all direct suppliers for the manufacture of Apple products

Where we're going

2023

2025

Goal
Transition
to 100% fiber-based packaging by 2025[20]

Goal
Use
100% recycled cobalt in all Apple-designed batteries[21]

100% recycled tin soldering and 100% recycled gold plating in all Apple-designed rigid and flexible printed circuit boards[22]

100% recycled rare earth elements in all magnets across all products[23]

2030

Apple 2030
To be carbon neutral for our entire footprint by 2030

Approach

Apple 2030

We have an ambitious commitment and a science-based plan to reach our Apple 2030 goal. We're focused on achieving reductions wherever possible, using approaches that offer clear evidence for a way forward while seeking to catalyze industry-wide change.

This begins with working to achieve carbon neutrality across our entire carbon footprint by 2030, setting ambitious targets to reduce our emissions by 75 percent. We prioritize carbon reductions, but for emissions that can't be mitigated using existing solutions we invest in high-quality carbon removal projects.

Our goal to be carbon neutral extends to our entire carbon footprint and is consistent with the Intergovernmental Panel on Climate Change's (IPCC) recommendation for global carbon neutrality.[24]

We're also committed to working toward reaching a 90 percent reduction in emissions from our 2015 baseline by 2050. Attaining deep decarbonization will require a collective, worldwide effort. Entire industries and economies must decarbonize. And while reaching a 90 percent reduction in emissions is outside Apple's or any one company's control, we're committed to taking actions that support this goal.

We've made significant progress by cutting emissions across our value chain by more than 55 percent since 2015. This started with our transition to sourcing 100 percent renewable electricity at our offices, retail stores, and data centers, which we achieved in 2018. And in 2020, we achieved carbon neutrality for our corporate emissions.[25]

APPLE 2030 ROADMAP

Addressing Apple's carbon footprint through four pillars

Design and materials

Designing products and manufacturing processes to be less carbon intensive through thoughtful material selection, increased material efficiency, greater product energy efficiency, the use of recycled and renewable materials in our products and packaging, and enhanced material recovery

Electricity

Increasing energy efficiency at our facilities and in our supply chain, and transitioning the electricity in our entire product value chain — including manufacturing and our customers' product use — to 100 percent clean electricity by 2030

Direct emissions

Reducing direct greenhouse gas emissions in our facilities and our supply chain through process innovation, emissions abatement, and shifting away from fossil fuels

Carbon removal

In parallel with our emissions reduction efforts, scaling up investments in carbon removal projects, including nature-based solutions that protect and restore ecosystems around the world

>75 percent emissions reduction

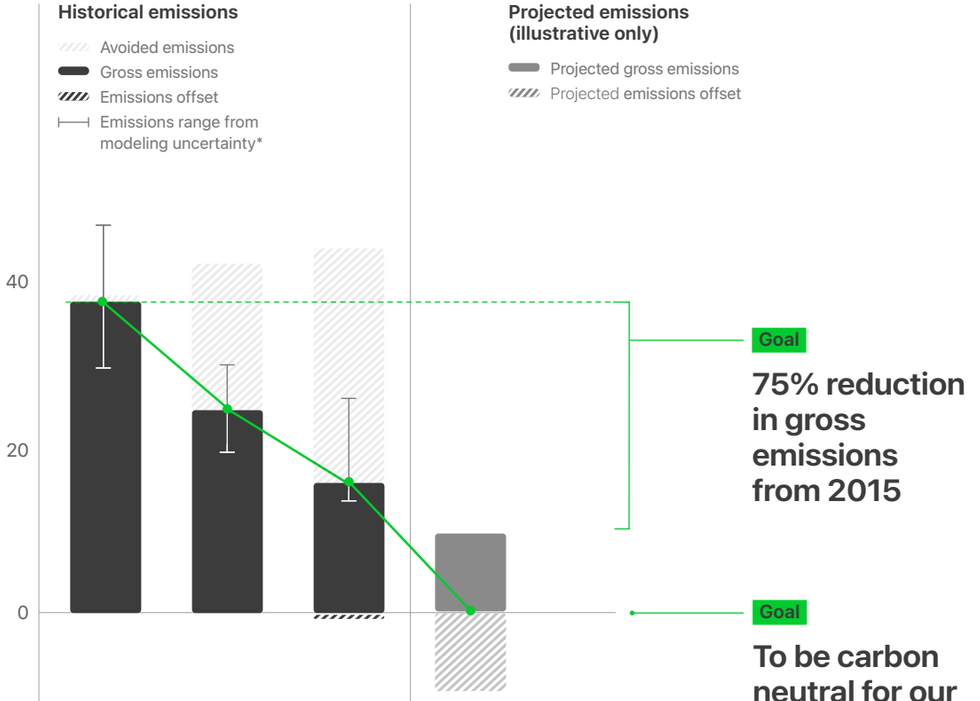
<25 percent of footprint

Environmental progress can and should be good for business. We underpin our climate strategy with strong business principles and innovation while harnessing the power of markets to replicate our solutions at scale — aiming to create the impact necessary to meet global reduction targets. We're also working to accelerate the global transition toward decarbonization while exploring the next generation of solutions and advocating for stronger policies. These efforts include fostering new and strengthened collaborations in public and private partnerships and investing in technological advancements. We've issued $4.7 billion in green bonds to model how businesses can drive investments to reduce global emissions.

Clear principles guide this work. Our emissions reduction targets are aligned with what current climate science shows is necessary to help limit warming to 1.5° C. We calculate our footprint across our value chain to include both direct and product-related emissions — from sourcing materials through end of life. We use the results of our detailed carbon accounting to adjust our Apple 2030 roadmap, which lays out our plan to become carbon neutral. And we're committed to disclosing our carbon footprint as well as our climate strategy and progress. This means sharing both challenges and successes. Our annual Environmental Progress Report, as well as our response to the global disclosure nonprofit CDP, provide details on our progress.

Transparency and disclosure are essential not only for sharing our climate strategy and progress but also for sending clear signals and inviting others to work with us. And progress needs to include low-income and historically marginalized communities that too often bear the brunt of climate change. We're pursuing ways to directly support these communities in our climate programs. To learn more about our work, read our feature on our Power for Impact program and our feature on our Impact Accelerator program.

APPLE'S PROGRESS TOWARD CARBON NEUTRALITY



Million metric tons CO_2e per fiscal year	2015	2019	2023	2030
Gross emissions	38.4	25.1	16.1	
Carbon offsets/removal	–	–	0.5	
Net emissions	38.4	25.1	15.6	

To see our progress toward our 2030 goal, see our Journey to Apple 2030 timeline.

In our latest Annual Green Bond Impact Report (PDF), we share progress on the projects funded in 2023.

Read our latest response to the CDP Climate Change 2023 questionnaire (PDF).

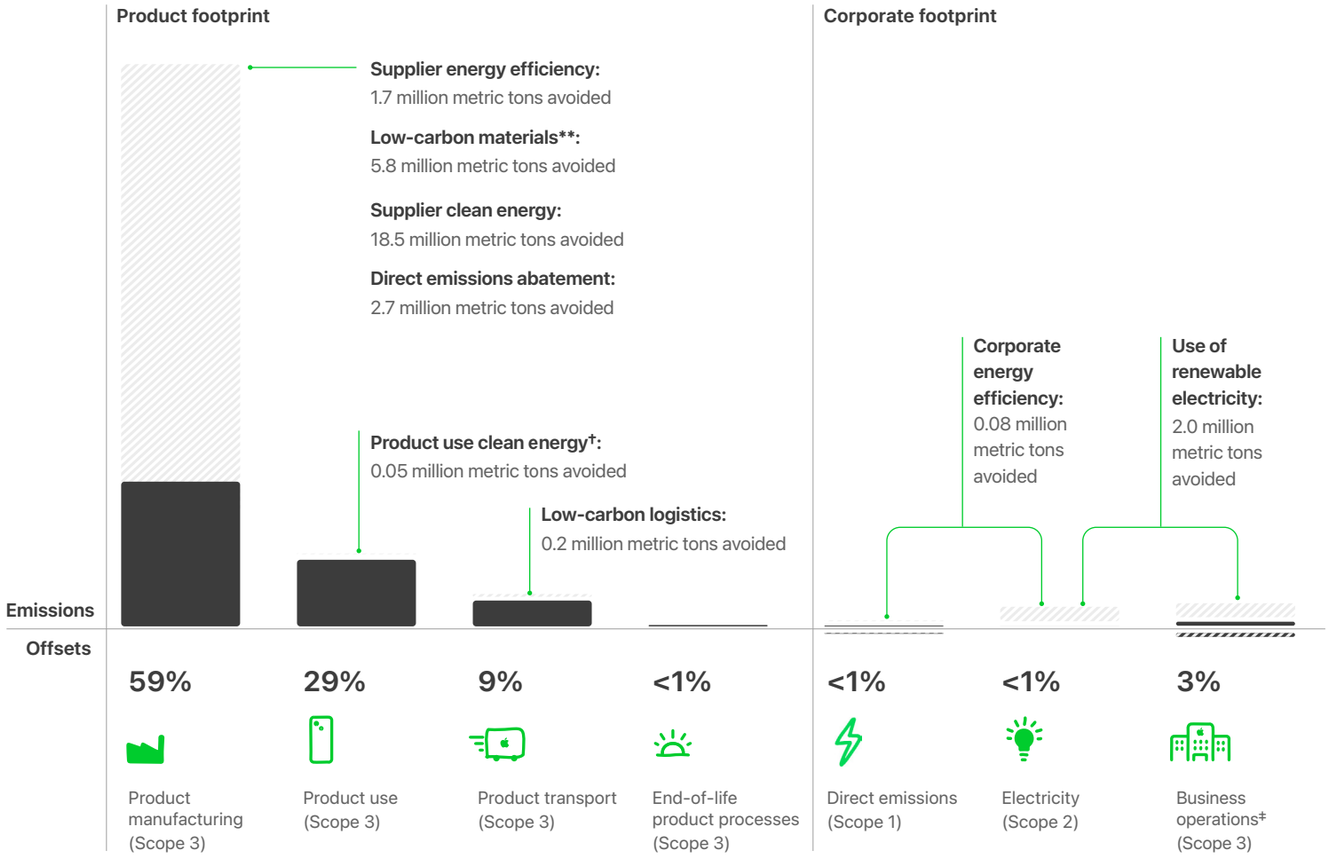
* Error bars: We're continuously refining our methodology to improve our carbon footprint estimate. But there's inherent uncertainty in modeling product-related greenhouse gas emissions, as illustrated by the error bars in this graphic.

Apple's comprehensive carbon footprint

In 2023, we estimate that our environmental programs avoided 31 million metric tons of emissions across all scopes. Initiatives that we've been growing for years continue to yield clear results, including sourcing 100 percent renewable energy for our facilities, transitioning suppliers to renewable energy, and using low-carbon materials in products.[26] While our revenue has grown by more than 64 percent since 2015, our gross emissions have decreased by more than 55 percent.

TOTAL EMISSIONS

15.6 Million Metric Tons Net Greenhouse Gas Emissions*



Avoided emissions
Gross emissions
Emissions offset

* **Net greenhouse gas emissions** represents our total gross footprint minus carbon offsets. Percentages shown for each emissions category represent the share of Apple's gross footprint. Totals add up to more than 100 percent due to rounding.

** **Low-carbon materials** represents emissions savings from transitioning to recycled materials in our products and using low-carbon aluminum, as described on page 33.

† **Clean energy** represents savings from clean energy procured by Apple or its suppliers.

‡ **Business operations** includes business travel, commute, work from home, upstream impacts from scope 1 fuels, and third-party cloud services.

Design and materials

We're designing our products to be less carbon intensive by prioritizing the use of recycled content and low-carbon materials and focusing on the energy efficiency of our software and hardware. We're working toward a future where every Apple product will be created from and contribute to circular supply chains. The design and material choices we make across our products will support reducing our carbon footprint.

Approach

Innovation drives our efforts around circularity — from the materials we source and the product design choices we make to the recycling and recovery innovations we pursue. We prioritize the materials and components that account for significant portions of our greenhouse gas emissions. This means that the choices we make product by product can scale toward reducing our overall footprint. These priorities inform our work to design for material efficiency and increase our use of recycled and renewable materials.

We aim to create products that use circular supply chains to one day end reliance on mined resources while meeting our rigorous standards for quality, durability, performance, and environmental and social protections. We strive for efficiency in sourcing and utilizing materials, relying on recycled and renewable materials for our products and packaging, and reducing scrap. And we also maintain strict standards for responsibly sourcing materials from primary, recycled, and renewable sources. Another part of what makes our goals possible is the recycling innovations we're developing to enhance material recovery. We hope our actions inspire others to support building circular supply chains.

Working to positively influence the markets where we work, advocating for policy that enables circular supply chains, impacting communities worldwide, and inspiring others to follow suit — these are the opportunities that drive us through the challenging work of creating circular supply chains.

Prioritizing less carbon-intensive materials and components
Focusing on those that account for significant portions of our greenhouse gas emissions

Designing products for material efficiency
Incorporating carbon emission profiles of the materials that go into our products

Increasing the use of recycled and renewable materials
Ensuring their use across our products

Driving product energy efficiency
Reducing energy use across our products

Design and materials addresses emissions from:

Product manufacturing (Scope 3)

Product use (Scope 3)

Product transport (Scope 3)

2023 progress

22%
recycled or renewable

Of the materials contained in products that were shipped to stores and customers, 22 percent came from recycled or renewable sources.

95%
recycled titanium

We introduced titanium to our priority materials list, and in our fall models of Apple Watch Ultra 2, when paired with Alpine Loop or Trail loop, we used 95 percent recycled titanium in the case.

58%
less energy

iMac uses 58 percent less energy than the ENERGY STAR requirement.*

* Energy consumption and energy efficiency values are based on the ENERGY STAR Program Requirements for Computers, including the max energy allowance for iMac. For more information, visit www.energystar.gov. ENERGY STAR and the ENERGY STAR mark are registered trademarks owned by the U.S. Environmental Protection Agency. For more information on the power consumption of iMac, read the Product Environmental Report.